

VISION & ARCHITECTURE

What we're building and how it fits together

BUILDCOST MVP | 48H HACKATHON | TIEMEN & GEORGES

1/3

THE MISSION

Upload a building plan

- ↓ AI extracts rooms & m²
- ↓ Discovers finishing level (QQPs)
- ↓ Calculates rebuild cost / m²

Target: Belgian building insurance companies

CORE FORMULA

Cost = m² × Base × ABEX × F

m²	Surface area from plan (WS1 — Tiemen)
Base	Price/m² by postcode (lookup table)
ABEX	Construction price index (provided file)
F	Finishing coefficient from QQPs (WS2 — Georges)

QQPs — THE HOLY GRAIL

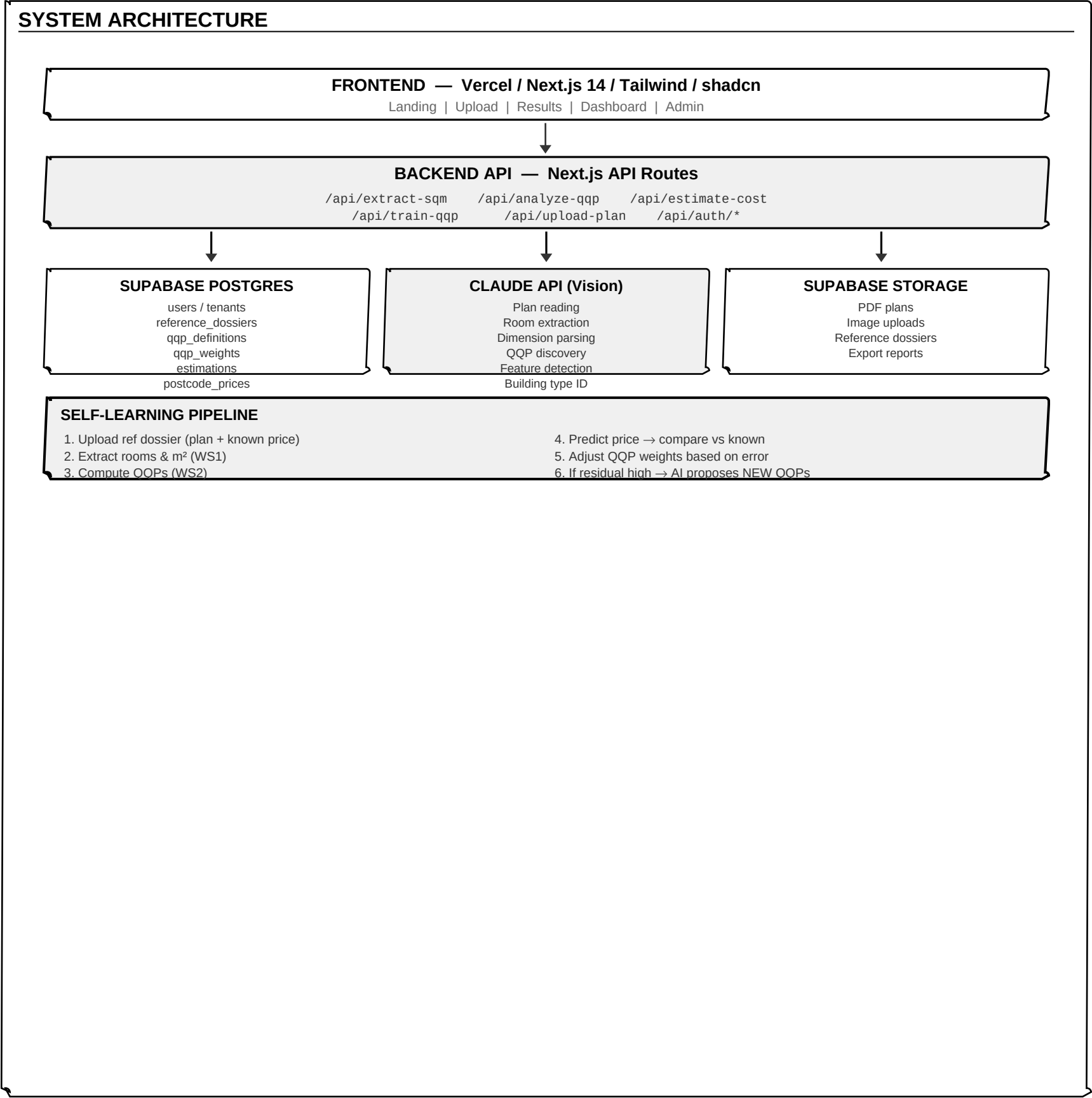
Quantitative-Qualitative Parameters that predict finishing level (and thus rebuild cost) from a plan.

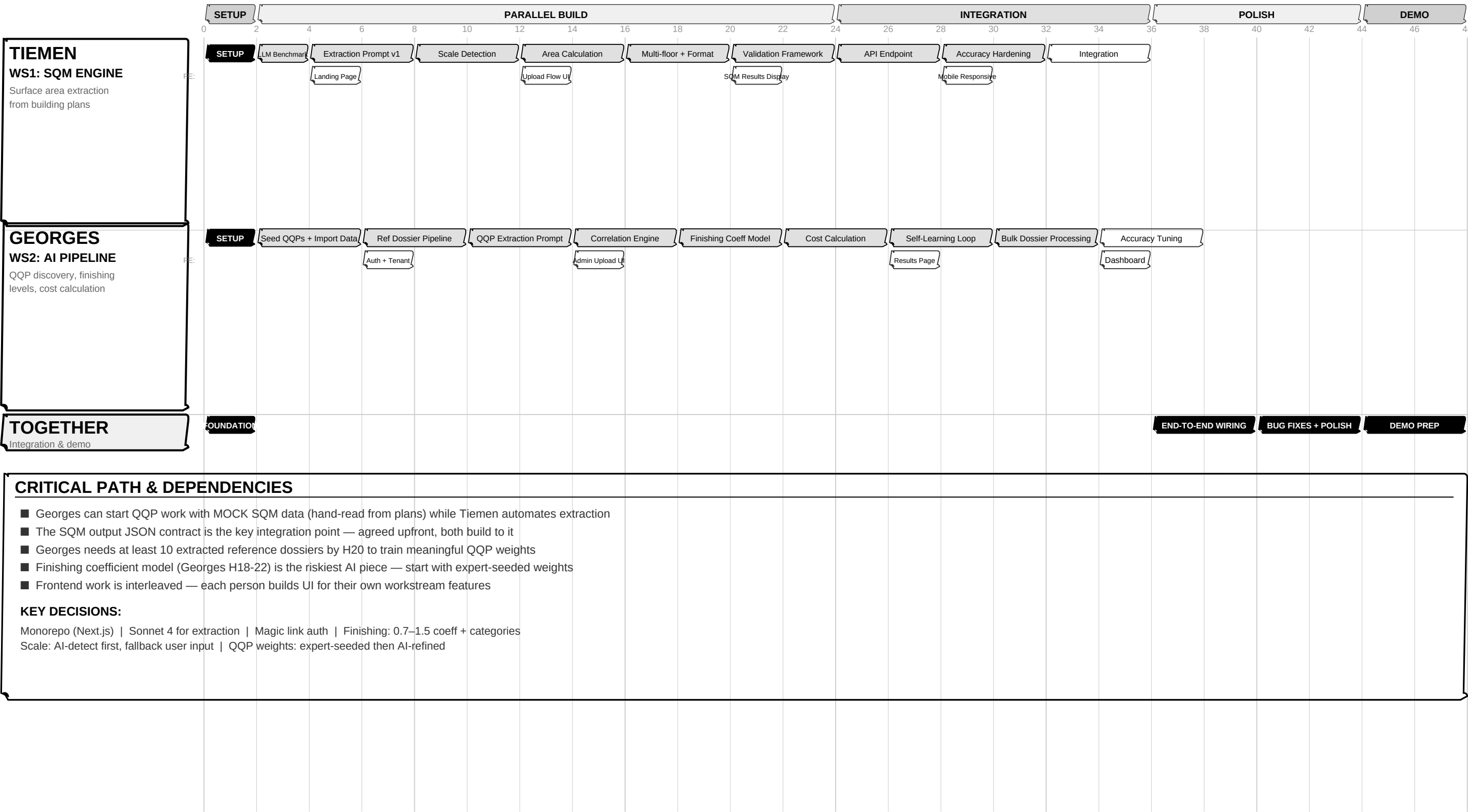
- Entrance hall size (m²)
- Nr of bathrooms + shower/bath combo
- Kitchen size & built-in appliances
- Living room area
- Materials visible on plan
- Garage / terrace / dressing / pool
- Nr of floors, ceiling height
- + AI discovers NEW ones from data!

Self-learning: 1000+ dossiers → discover & weight QQPs

!! NOT REAL ESTATE VALUATION !!

This is REBUILD COST. A luxury apt costs the same to rebuild in Knokke or Charleroi. Postcode coeff is separate.





WORKSTREAMS & CHECKLISTS

Detailed scope, outputs, and checkboxes for each track

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3/3

TIEMEN — WS1: SQM ENGINE

Mission: Plan upload → precise room-by-room m²

CORE SQM:

- | | |
|---|--------|
| ☐ Benchmark LLM vision APIs on 5 real plans | H2-4 |
| ☐ Build extraction prompt v1 (rooms, dims, labels) | H4-8 |
| ☐ Scale detection (bar / door-width / dim text) | H8-12 |
| ☐ Area calculation engine (irregular rooms) | H12-16 |
| ☐ Multi-floor + multi-format (PDF, images) | H16-20 |
| ☐ Validation: extracted m² vs known from refs | H20-24 |
| ☐ /api/extract-sqm endpoint ready | H24-28 |
| ☐ Accuracy hardening (fix worst cases) | H28-32 |

FRONTEND:

- | | |
|--|--------|
| ☐ Landing page (hero, value prop, CTA) | H4-6 |
| ☐ Upload flow UI (drag & drop + postcode input) | H12-14 |
| ☐ SQM results display (room table, floor summary) | H20-22 |
| ☐ Mobile responsive pass | H28-30 |

GEORGES — WS2: AI PIPELINE

Mission: QQPs → finishing level → rebuild cost

CORE AI:

- | | |
|--|--------|
| ☐ Seed 20-30 QQP definitions in Supabase | H2-4 |
| ☐ Import postcode price table + ABEX index | H4-6 |
| ☐ Reference dossier pipeline (upload + store) | H6-10 |
| ☐ QQP extraction prompt (from plan data) | H10-14 |
| ☐ Correlation engine (QQP vs price analysis) | H14-18 |
| ☐ Finishing coefficient model (0.7 to 1.5) | H18-22 |
| ☐ /api/estimate-cost endpoint (full formula) | H22-26 |
| ☐ Self-learning loop (weight updates) | H26-30 |
| ☐ Process 50+ reference dossiers | H30-34 |

FRONTEND:

- | | |
|--|--------|
| ☐ Supabase Auth + multi-tenant setup | H6-8 |
| ☐ Admin: reference dossier upload UI | H14-16 |
| ☐ Results page + cost breakdown | H26-28 |
| ☐ Dashboard (history, accuracy, QQPs) | H34-36 |

SQM OUTPUT CONTRACT (WS1 → WS2)

```
{  "plan_id": "uuid",  "floors": [{    "level": 0,    "label": "Gelijkvloers",    "rooms": [{      "label": "Living",      "area_sqm": 42.5,      "category": "living|bedroom|bathroom|",      "features": ["fireplace", "large_window"],      "confidence": 0.92    }  ]  },  "total_sqm": 145.3}, {"summary": {  "total_livable_sqm": 245.6,  "total_gross_sqm": 308.8,  "floor_count": 2 }}
```

SUCCESS CRITERIA & RISKS

MUST HAVE FOR DEMO:

- ☐ Upload plan → room-by-room m² (>80% accuracy)
- ☐ Building type auto-detected
- ☐ Finishing level from QQPs → coefficient
- ☐ Rebuild cost calculated and displayed
- ☐ 50+ reference dossiers processed for training
- ☐ Auth + multi-tenant + mobile-friendly

TOP RISKS:

- Plan quality varies → multi-prompt fallback
- Scale not on plan → door-width calibration
- Claude hallucination → show extraction for validation
- Not enough dossiers in 48h → export seeded weights

FINISHING LEVELS (Coefficient Range)

BASIC 0.70–0.85	STANDARD 0.85–1.00	COMFORT 1.00–1.15	LUXURY 1.15–1.35	PREMIUM 1.35–1.50
Simple materials, minimal equipment	Average Belgian new build	Quality finishes, good equipment	High-end materials, full equipment	Exceptional quality, custom everything